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# How to Remove a Public IP from an AWS EC2 Instance without Restart or Termination



Shane Fast

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4 min read · Jun 23, 2020



134



10



boop

I was doing some security upgrades when I ran into a particularly novel problem — once an instance is started, there isn't a magical button in AWS to remove an auto-assigned public IP. Certainly, there had to be others out there who have faced a similar issue. After all, removing the public IP to a server is a surefire way to help prevent unwanted mischief from the world at large.

Of course, you could always take a snapshot, terminate the original instance, and start a new instance without a public IP, but that takes time. There's got to be a better way, right? Well, after taking a look at what other people suggested, I was disheartened initially. Let's see some forums weigh in on the topic:

2

4 years ago

### remove public ip from instance

I'm trying to convert some of my subnets from public to private. I have set a new public subnet and created a nat gateway inside this. On the public subnet i wish to turn private, I've turned off auto assign IP and changed the route table to default to the NAT gateway.

The instance inside this subnet still has a public ip address. How do i remove this ?

Thanks,

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3 points · 4 years ago

You can't manually remove the public IP address from an instance after launch. You'll have to relaunch in the private subnet.

<http://docs.aws.amazon.com/AmazonVPC/latest/UserGuide/vpc-ip-addressing.html#subnet-public-ip>

2 points · 4 years ago

You're going to have to relaunch the instance. Public IPs are only assigned (or not assigned) at launch.

2 points · 4 years ago

To give a little more detail/clarity. The answer is you can't remove them.

However you can create an AMI and re-create the instance, from the AMI, without the public IP.

[https://www.reddit.com/r/aws/comments/4ocpd8/remove\\_public\\_ip\\_from\\_instance/](https://www.reddit.com/r/aws/comments/4ocpd8/remove_public_ip_from_instance/)

Can I remove the public IP on my instance without terminating it?

0

votes

1 view

asked Jul 16, 2019 in AWS by (19.2k points)

I have several instances on a vpc that communicate with each other through their private IPs. Each instance was launched some time ago and assigned a random public IP which is not used for anything. Since anything that is public presents a vulnerability and they are not used at all, I would like to remove them.

Is it possible to Remove public Ip Addresses without terminating the instance?

amazon-ec2

amazon-web-services

answer

1 Answer

0

votes

answered Jul 16, 2019 by (44.6k points)

If you are using an Elastic IP then it would be possible. Because when you remove it, then the normal public IP address will be given. Or you can create an image of the instance, and create another EC2 instance using it which will surely have a different IP.

comment

<https://intellipaat.com/community/11018/can-i-remove-the-public-ip-on-my-instance-without-terminating-it>

<https://medium.com/bacic/how-to-remove-a-public-ip-from-an-aws-ec2-instance-without-restart-or-termination-c91c22e1ce3f>

2/10

Ok, not looking good. What about the AWS forum?

### Removing an VPC instance public IP address

Posted by [REDACTED]

Posted on: Jun 19, 2015 1:52 PM

★ This question is **answered**.

Hello,

I launched an instance in a VPC and I set the 'auto-assign public ip' setting to enable. Now I would like the public IP removed (disabled) but can't see where in the console/cli this setting is stored. Every time I start the instance it gets a public IP. How can I change the default to disable?

Thanks

Replies: 1 | Pages: 1 - Last Post: Jun 20, 2015 1:43 PM by: GMan

## Replies

### Re: Removing an VPC instance public IP address

Posted by: [REDACTED]

Posted on: Jun 20, 2015 1:40 PM

in response to: [REDACTED]

★ Helpful

Hi,

The configuration of whether or not to assign a public IP address by default, is controlled by the VPC subnet setting. You may modify the subnet to not auto-assign public IP's when launching instances into that subnet.

Please see Link: <http://docs.aws.amazon.com/AmazonVPC/latest/UserGuide/vpc-ip-addressing.html#subnet-public-ip> for more details on how to change the configuration of your VPC.

**Note** You can't manually disassociate the public IP address from your instance after launch. Instead, it's automatically released in certain cases, after which you cannot reuse it. See Link: <http://docs.aws.amazon.com/AmazonVPC/latest/UserGuide/vpc-ip-addressing.html#vpc-public-ip>

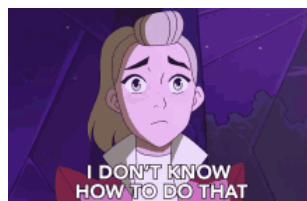
If you found this **helpful** please mark the question as such, I'm trying to up my forum ranking.

Edited by: GMan on Jun 20, 2015 1:42 PM

<https://forums.aws.amazon.com/thread.jspa?threadID=185690>

Ouch! Well looks like it can't be done...

. . .



. . .

...But that's not the way we do it! Who the hell do you think I am? (Sorry about the obscure reference 😂)

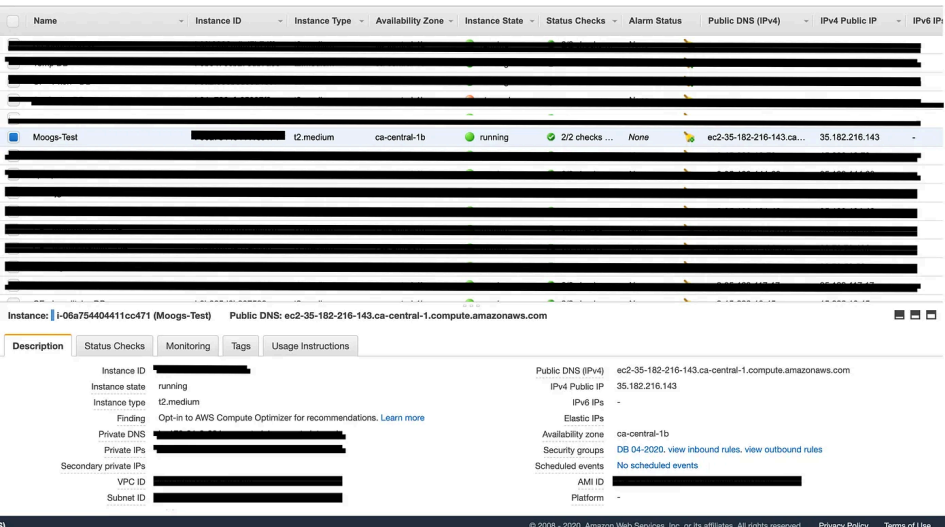


After a bit of struggle, here is the step-by-step process of removing an auto-assigned public IP from an AWS EC2 instance:

Instructions

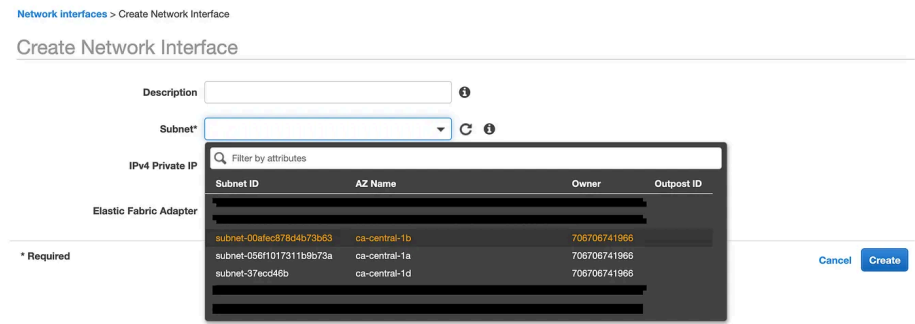
Be sure to follow these steps in this exact order!

- 1. Go to your running EC2 instance with the public IP you want to remove. Note what region and availability zone are associated with that instance.



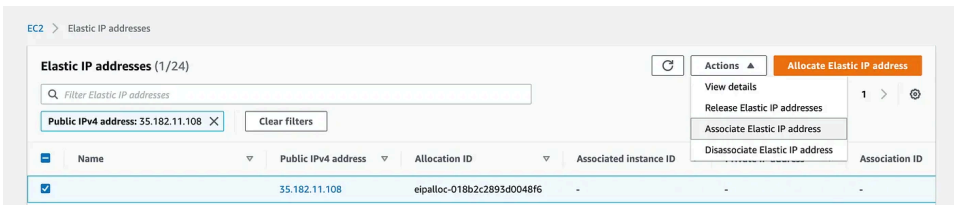
In this example, we are using an instance in the Canadian central region in the 1b availability zone

- 2. On the left-hand panel, scroll and go into “Network Interfaces” under the “Network and Security” Dropdown. Once there, click “Create Network Interface.” In the description, give it a memorable name (we used “TEST 1b” in this example). For the subnet, pick a matching one for the instance from Step 1. Click “Create” to finish.

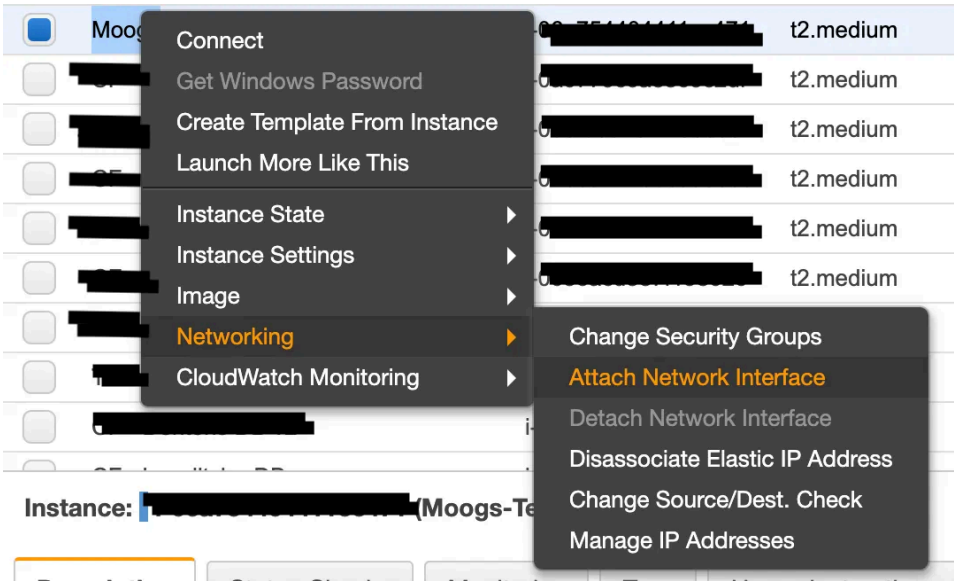


ca-central-1b in this example

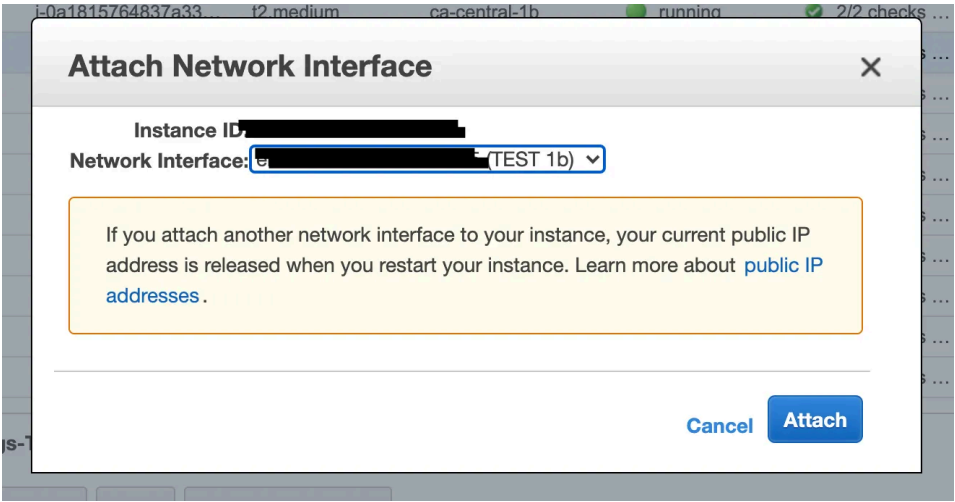
3. Next, go it elastic IPs (also under the “Network and Security” Dropdown). After you create a new elastic IP, associate it with the EC2 instance.



4. Next, go back to your running EC2. Right-click it and select “Attach Network Interface” under the “Networking” option



5. From here, attach the network interface made previously.



6. Next, disassociate the elastic IP from the instance. If done properly, the public IP should disappear.

7. Finally, detach the network interface, this will remove the second private IP that was added.



*Make sure to disassociate the elastic IP first before detaching the extra network interface!*

. . .

Overall this process takes only about 30 seconds as opposed to potentially an hour or more to take a snapshot and restart a new instance (not including any specific configuration that might be needed afterward!). The elastic IP and network interface can be used over and over again if you have multiple instances, you need to do this in the same availability zone. I hope you find this helpful!



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It's easy to lose sight of what's important in life. As developers and people deeply interested in technology, we can easily forget why we started this pursuit. These are our projects, discoveries, and stories we want to carry into the future to remember and share with the world.



## Written by Shane Fast

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Interested in building things and building teams.

## Responses (10)



Chris Lindgren

What are your thoughts?



Zdravko Posloncec  
Dec 11, 2020



Thank you! I've created a bash script based on your document. <https://github.com/zposloncec/ec2-public-ip>



7



1 reply

[Reply](#)



Tarun Theogaraj  
Jan 25, 2024



I tried this and it works. Is there a way to do it in reverse? I have an EC2 instance that was launched without auto-assigning a public IPv4 address (it does have an IPv6 address). Is there some way to assign a public (non elastic IP) IPv4 address after the instance has already been launched?



1

[Reply](#)



Keshav Chhaparia  
Dec 7, 2024



this worked..thank you!!



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See all responses

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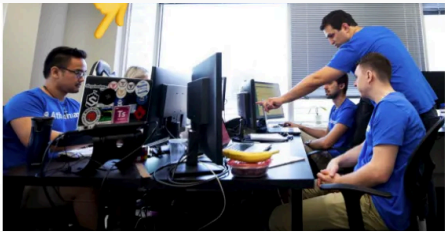


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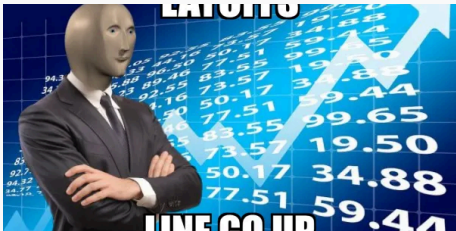


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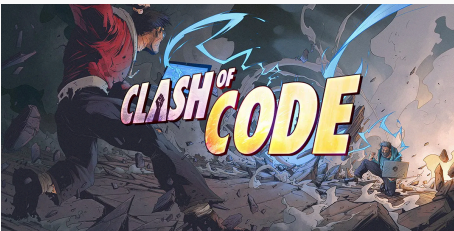


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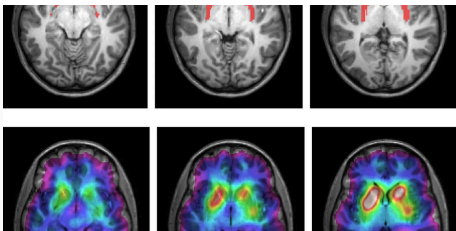
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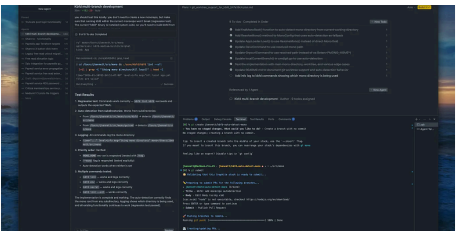


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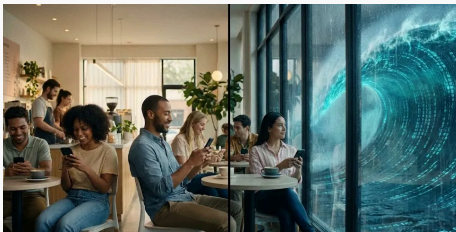


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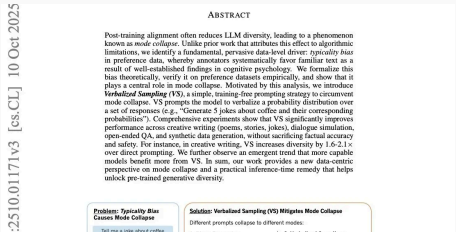


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